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ΠΑΝΕΠΙΣΤΗΜΙΟ
ΔΥΤΙΚΗΣ ΑΤΤΙΚΗΣ
UNIVERSITY OF WEST ATTICA

DEPARTMENT OF COMPUTER AND INFORMATION
TECHNOLOGY AND COMPUTER ENGINEERING

LAB 1
RESISTANCE CODES - MEASURING INSTRUMENTS -
OSCILLOSCOPE - AMPLITUDE AND FREQUENCY
MEASUREMENT

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DELIVERY DATE : 4/11/2022

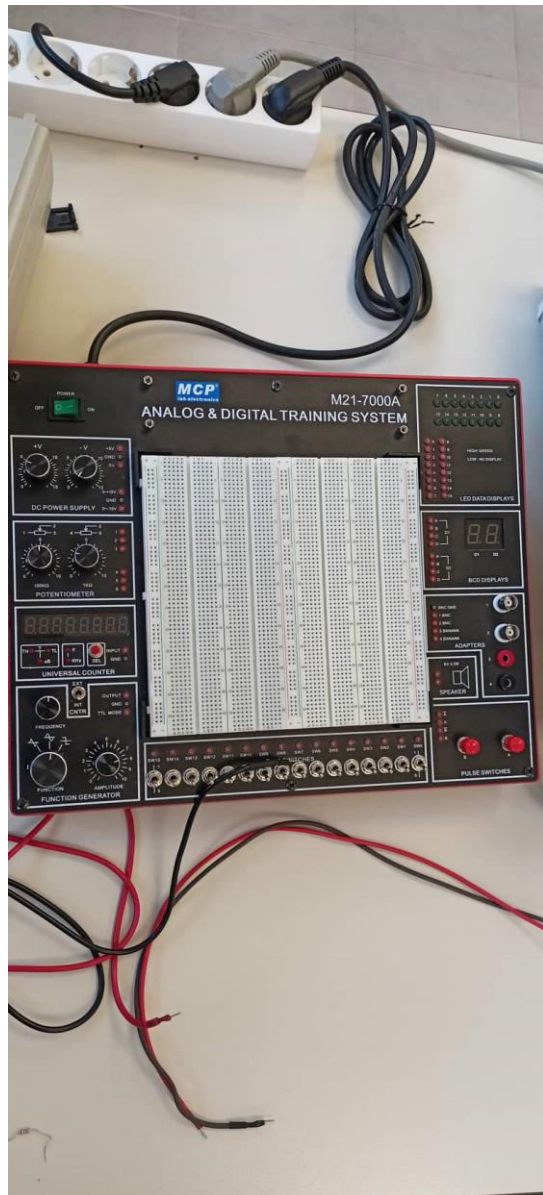
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STUDENT PHOTO :

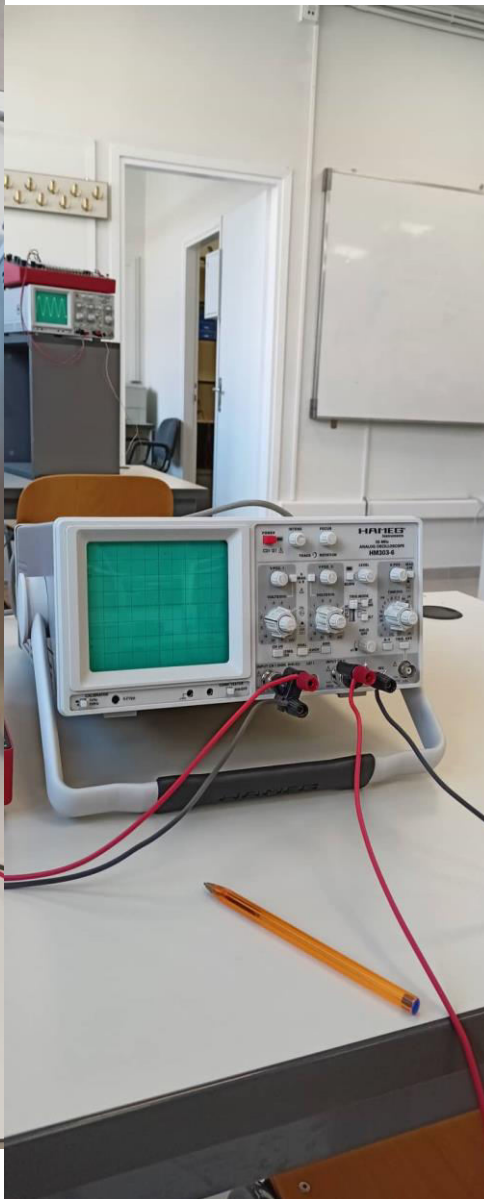


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PHOTOGRAPHS OF EQUIPMENT USED IN THE LABORATORY



Breadboard M21-7000A



Digital HM203-5 Multimeter MT8045



Oscilloscope HAMEG

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Resistor 1 KOhm



Resistor 27 KOhm



Resistor 39 KOhm



Cable

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1. Resistor Resistance Measurement

General

In the chapters "Simulated Solution" and "Experimental Solution" the components used and the observations made during the experiment are presented in detail with snapshots from the "Multisim" software and photos of the experiment from the laboratory environment respectively, while in the chapter "Theoretical Solution" the mathematical proof of the experiment is analysed.

Theoretical Solving

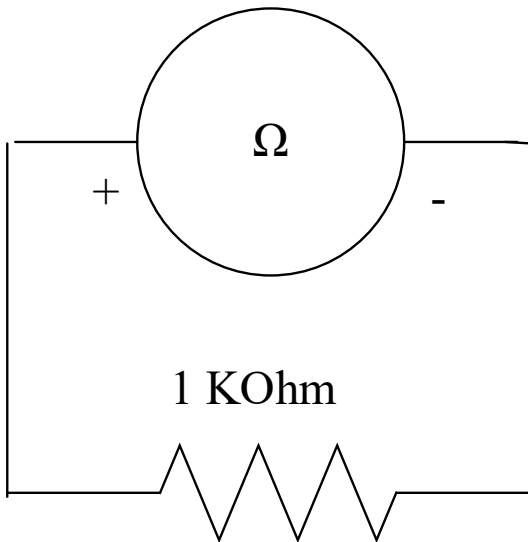
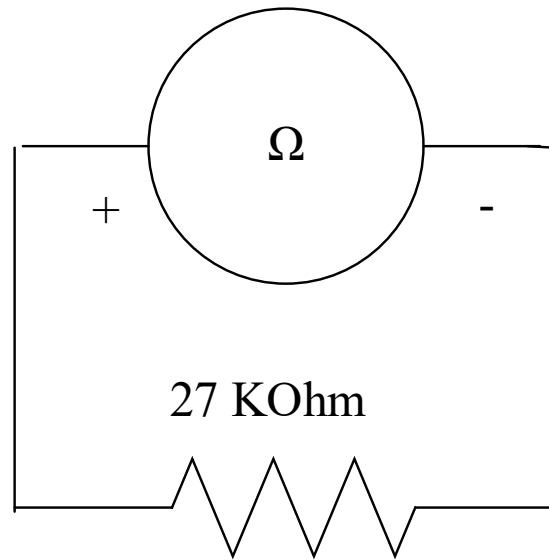


Figure 1



Picture 2

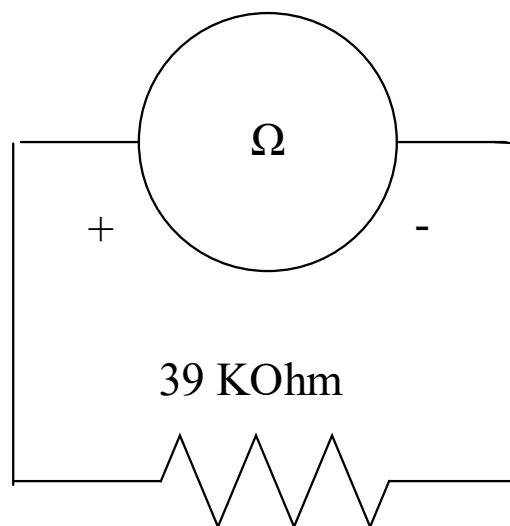


Figure 3

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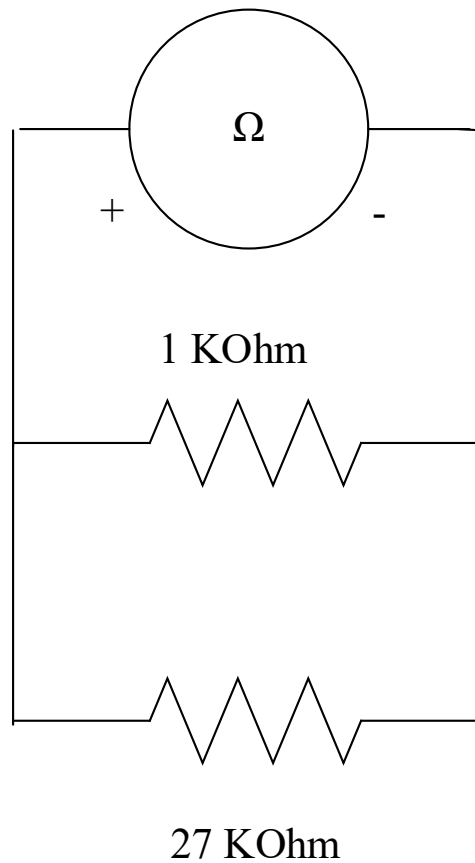


Figure 4

Figures 1, 2, 3 and 4 show the schematic illustration of the resistance measurement of three different resistors. Specifically, the measurement is performed on resistors with a resistance of 1 KOhm (Figure 1), 27 KOhm (Figure 2), 39 KOhm (Figure 3) and 0.97 KOhm respectively (Figure 4). In Figure 4, the two resistors with a resistance of 1 KOhm and 27 KOhm are connected in parallel and therefore the equivalent resistance is :

$$\frac{1}{R_{O\Lambda}} = \frac{1}{1 \times 10^3} + \frac{1}{27 \times 10^3} \rightarrow R_{O\Lambda} = \frac{1 \times 10^3 \times 27 \times 10^3}{1 \times 10^3 + 27 \times 10^3} \rightarrow R_{O\Lambda} = \frac{27 \times 10^6}{28 \times 10^3} \rightarrow$$
$$R_{O\Lambda} = 0.97 \text{ KOhm}$$

The measurement can be carried out using an ohmmeter or multimeter set to calculate resistance.

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Simulative Resolution

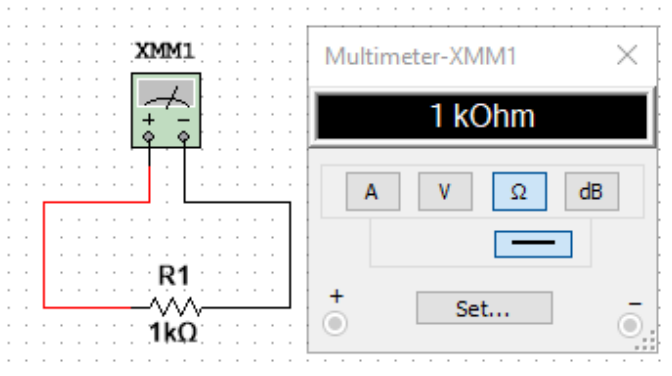
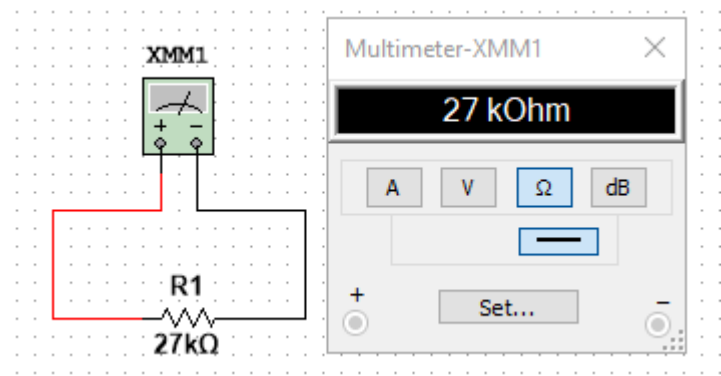


Figure 4



Picture 5

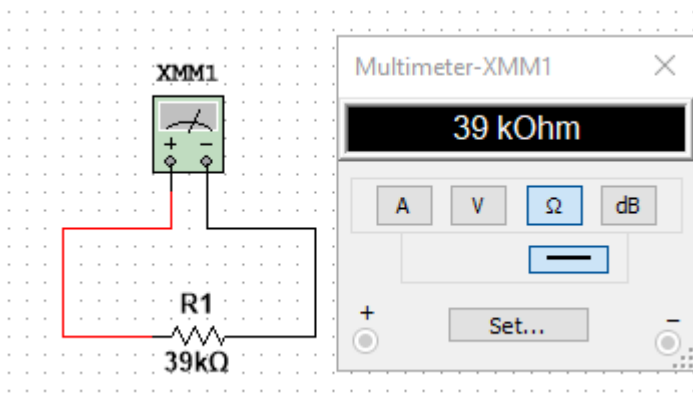
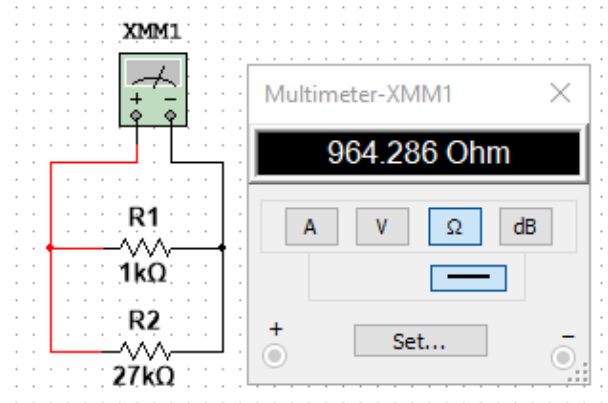


Figure 6



Picture 7

From the simulated solution of the experiment in the simulation software "Multisim" (Figures 4, 5, 6, 7), the ideal resistance values of 1 KOhm (Figure 4), 27 KOhm (Figure 5), 39 KOhm (Figure 6), and the equivalent resistance of 1 KOhm to 27 KOhm connected in parallel (Figure 7) are clearly illustrated. The measurement is performed using an analog multimeter set to calculate resistance and connected in parallel to the ends of each resistor. It is observed that the equivalent resistance from the multimeter reading in Figure 7, is 964.286 Ohm or 0.97 KOhm, which verifies the mathematical proof in "Theoretical Solution" (pages 6-7).

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Experimental Solution

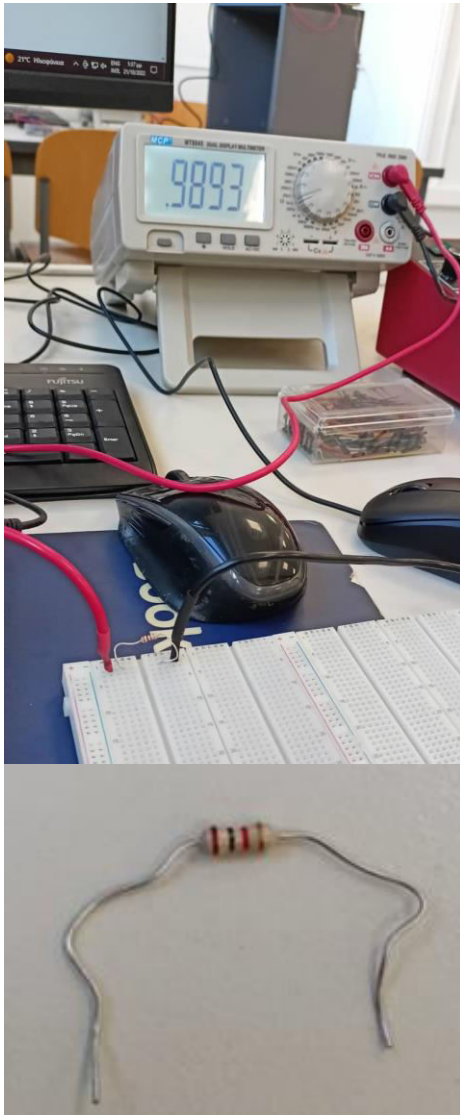


Figure 8

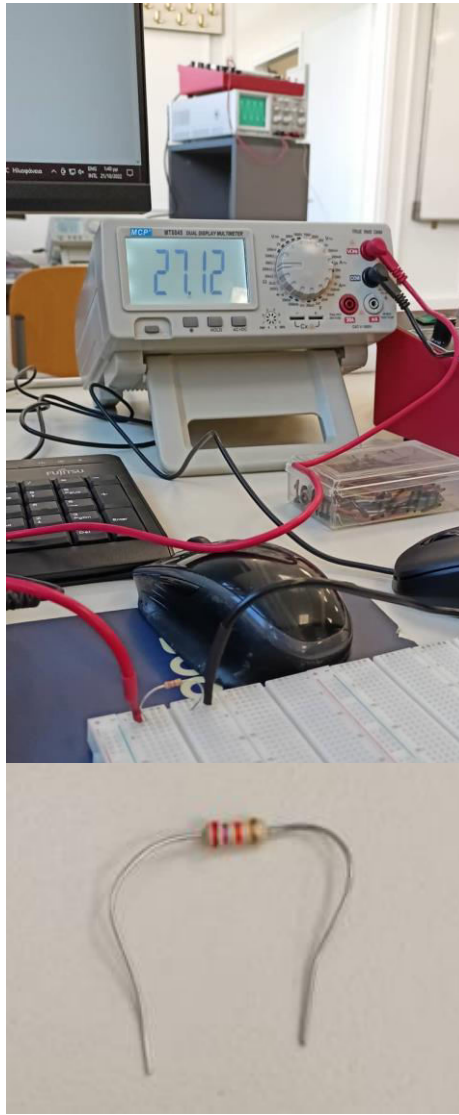


Figure 9

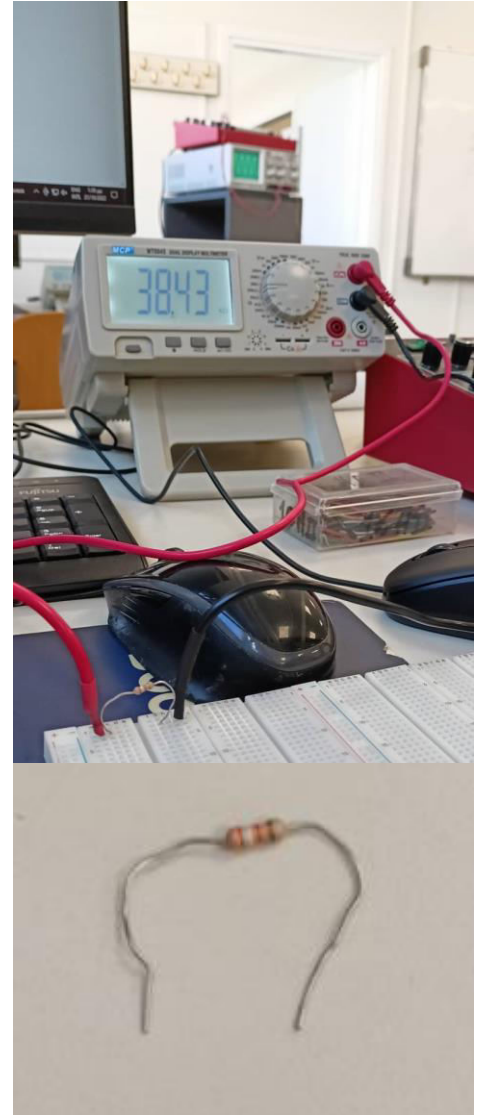


Figure 10

Resistance 1 KOhm

For the experimental solution carried out in the laboratory environment, the interface board "Breadboard M21-7000A", a digital multimeter bench "MT8045", three resistors of resistance 1 KOhm, 27 KOhm and 39 KOhm were used.

First, we connect the ends of the resistor with a resistance of 1 KOhm on the board, to holes that are not the same point, where the same point is a horizontal five-hole pattern (Figure 8). Then, after setting the digital multimeter to measure resistance, we connect the terminals parallel to the ends of the resistor. We place the positive terminal (red wire) on the left and the negative terminal (black wire) on the right, set the scale to the next highest of the resistance of the resistor we are measuring, i.e., 2 KOhm, and notice that the multimeter reading in Figure 8 is 0.9893 KOhm, not 1 KOhm. This is because the resistor has a tolerance of -5%, i.e. -5%

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deviation from its true value. According to the color code of resistors, the gold 4th stripe on the resistor indicates a $\pm 5\%$ tolerance. Table 1 details the colour code of the 1 KOhm resistor.

Color	1 st strip	2 nd strip	3 rd strip	4 th strip
Coffee	1			
Black		0		
Red			10^2	
Gold				$\pm 5\%$

Table 1

Resistance 27 KOhm

The procedure is the same for the other resistors (Figure 9, Figure 10). In more detail, in Figure 9 the multimeter reading for the 27 KOhm resistor is 27.12 KOhm (the scale is set to 200 KOhm). This is because the resistor has a tolerance of $\pm 5\%$, i.e. $\pm 5\%$ deviation from its true value. According to the colour code of resistors, the gold 4th stripe on the resistor marks a tolerance of $\pm 5\%$ tolerance. Table 2 details the colour code of the 27 KOhm resistor.

Color	1 st strip	2 nd strip	3 rd strip	4 th strip
Red	2			
Purple		7		
Orange			10^3	
Gold				$\pm 5\%$

Table 2

Resistance 39 KOhm

In Figure 10 the multimeter reading for the 39 KOhm resistor is 38.43 KOhm (the scale is set to 200 KOhm). This is because the resistor has a tolerance of $\pm 5\%$, i.e. $\pm 5\%$ deviation from its true value. According to the color code of resistors, the gold 4th stripe on the resistor marks a $\pm 5\%$ tolerance. Table 3 details the colour code of the 39 KOhm resistor.

Color	1 st strip	2 nd strip	3 rd strip	4 th strip
Orange	3			
White		9		
Orange			10^3	
Gold				$\pm 5\%$

Table 3

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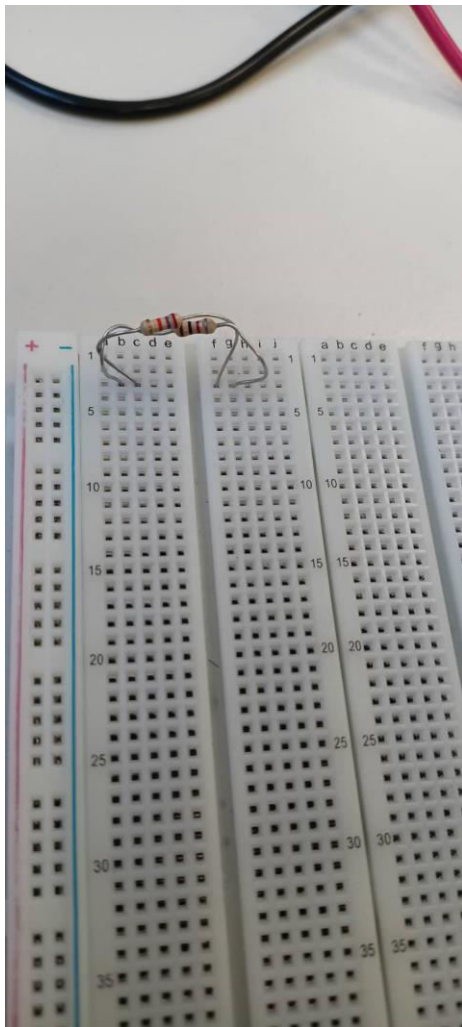


Figure 11

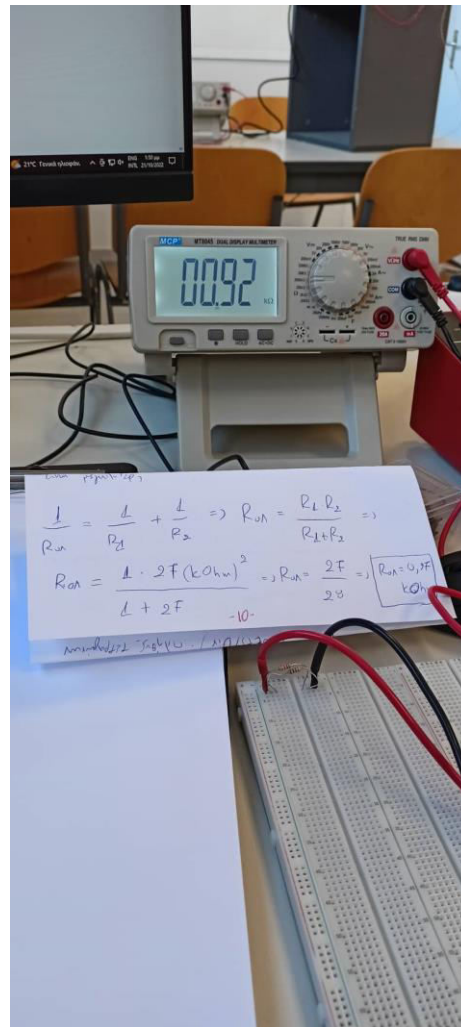


Figure 12

Resistance 0.97 KOhm

In Figure 11, the two resistors of 1 KOhm and 27 KOhm are connected in parallel. In Figure 12, the resistance measurement procedure is the same as the previous resistor measurements (the scale is set to 2 KOhm). The reading of the instrument is 0.92 KOhm, a value that is far from the actual value of the equivalent resistance (0.97 KOhm) due to the tolerances of the two resistors.

2. DC Voltage Measurement D.C.

General

In the chapters "Simulated Solution" and "Experimental Solution" the components used and the observations made during the experiment are presented in detail with snapshots from the "Multisim" software and photos of the experiment from the laboratory environment respectively, while in the chapter "Theoretical Solution" the mathematical proof of the experiment is analysed.

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Theoretical Solving

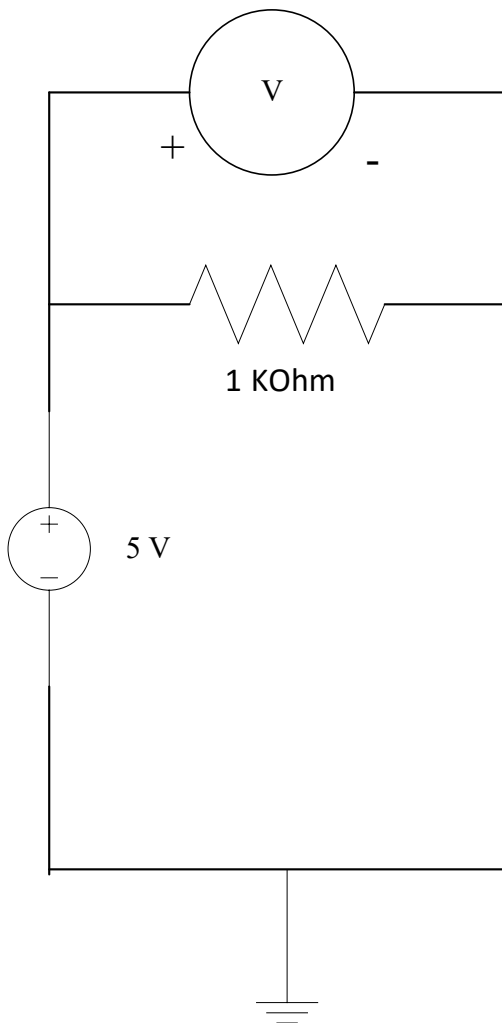


Figure 13

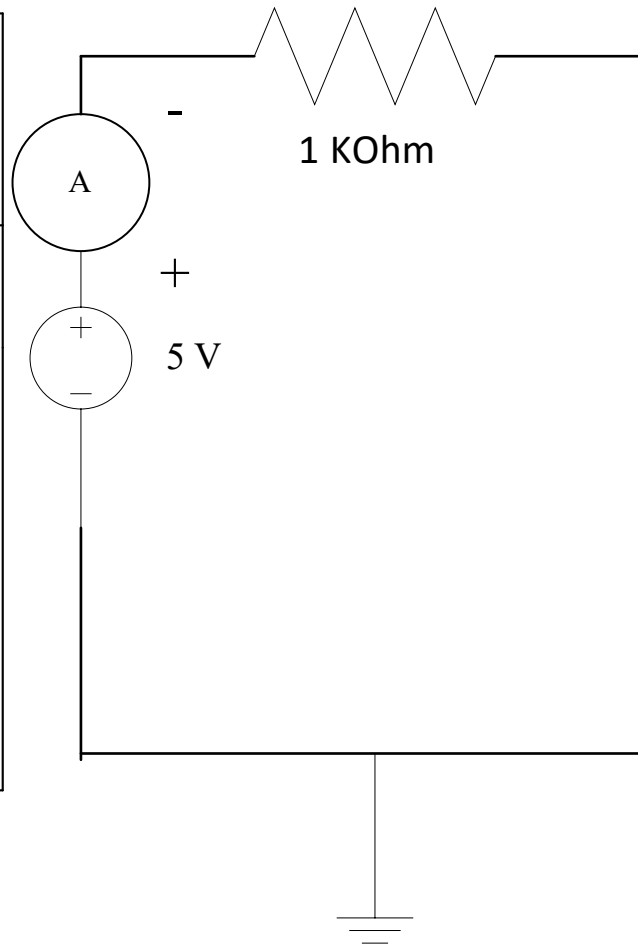


Figure 14

Figures 13 and 14 show the schematic of a simple circuit with a DC voltage source equal to 5 Volts. A resistor of 1 KOhm is also connected to the circuit, and with the help of a voltmeter, the potential difference at the ends of the resistor is calculated (Figure 13), and with the help of an ammeter, the intensity of the current flowing through the resistor is calculated (Figure 14). The potential difference at the ends of the resistor is the same as the source voltage, because by Kirchhoff's law of voltages, the sum of the potential differences in the branches of a loop equals 0.

$$V_{\text{source}} + V_{\text{ground}} + V_{\text{resistor}} = 0 \rightarrow 5 \text{ Volt} + 0 \text{ Volt} - 5 \text{ Volt} = 0 \rightarrow 0 = 0$$

The intensity of the current flowing through the resistor is calculated using Ohm's law, where the potential difference at the ends of the resistor is linearly proportional to the intensity of the current flowing through it.

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$$I = \frac{V_{\text{source}}}{R} \rightarrow I = \frac{5 \text{ Volt}}{1 \times 10^3 \text{ Ohm}} \rightarrow I = 5 \times 10^{-3} \text{ A} \rightarrow I = 5 \text{ mA}$$

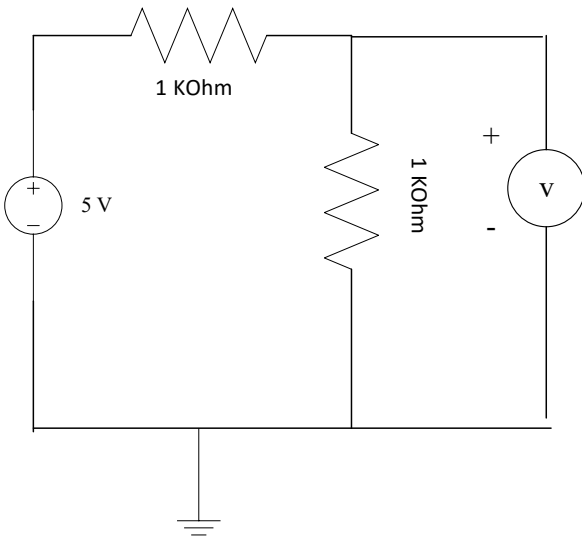


Figure 15

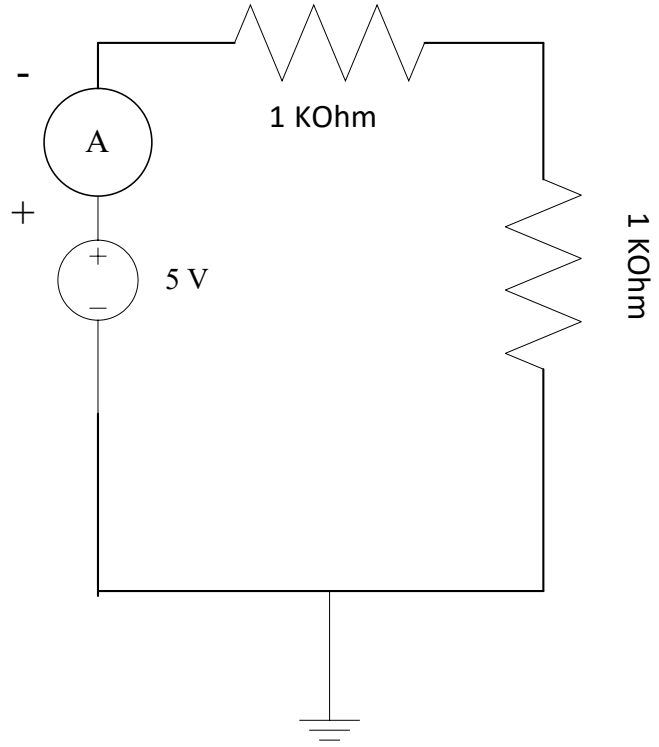


Figure 16

Figures 15 and 16 show the schematic of a simple circuit with a DC voltage source equal to 5 Volts. Two resistors of 1 KOhm are also connected to the circuit in equivalent series and the potential difference between one end of one resistor and one end of the other resistor is calculated with the help of a voltmeter (Figure 15), and the intensity of the current flowing through the resistors is calculated with the help of an ammeter (Figure 16). The potential difference at the ends of the resistors, which is also the same due to equivalent resistance, is calculated from the voltage divider formula.

$$V_{1 \text{ KOhm}} = \frac{R_{1 \text{ KOhm}}}{R_{1 \text{ KOhm}} + R_{1 \text{ KOhm}}} \times V_{\pi\eta\gamma\eta} \rightarrow V_{1 \text{ KOhm}} = \frac{1 \times 10^3}{1 \times 10^3 + 1 \times 10^3} \times 5 \text{ Volt} \rightarrow$$

$$V_{1 \text{ KOhm}} = 2.5 \text{ Volt}$$

The intensity of the current flowing through the resistors is calculated using Ohm's law, where the potential difference at the ends of the resistor is linearly proportional to the intensity of the current flowing through it. The equivalent resistance is calculated from the resistor series connection formula.

$$R_{I\Sigma} = R_{1 \text{ KOhm}} + R_{1 \text{ KOhm}} \rightarrow R_{I\Sigma} = 1 \times 10^3 \text{ KOhm} + 1 \times 10^3 \text{ KOhm} \rightarrow R_{I\Sigma} = 2 \text{ KOhm}$$

Ohm's Law :

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$$I = \frac{V_{\text{source}}}{R_{EQ}} \rightarrow I = \frac{5 \text{ Volt}}{2 \times 10^3 \text{ Ohm}} \rightarrow I = 2.5 \times 10^{-3} \text{ A} \rightarrow I = 2.5 \text{ mA}$$

Simulative Resolution

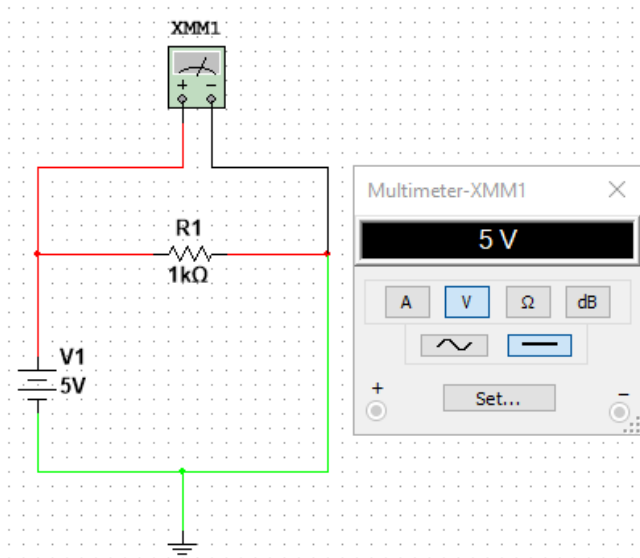


Figure 17

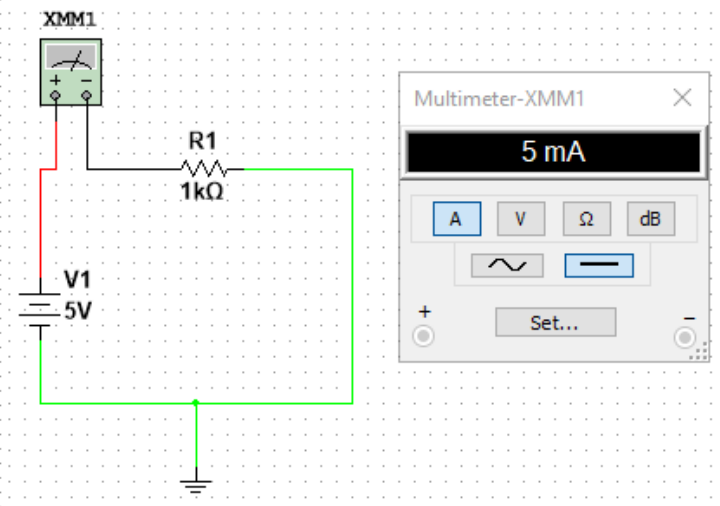


Figure 18

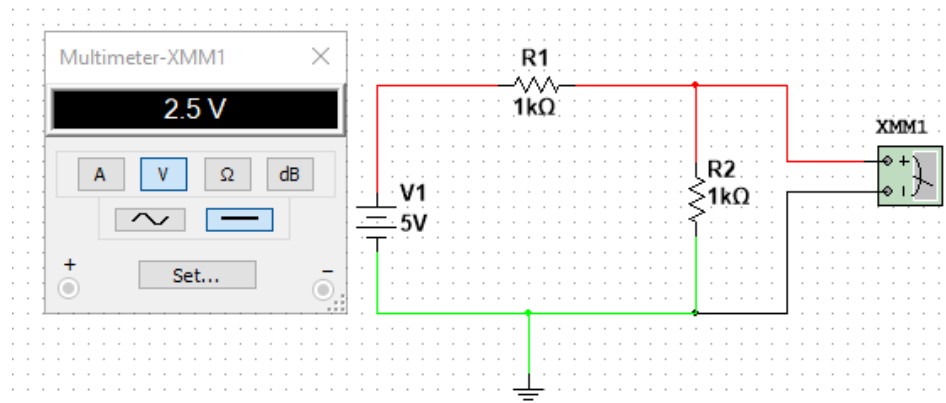


Figure 19

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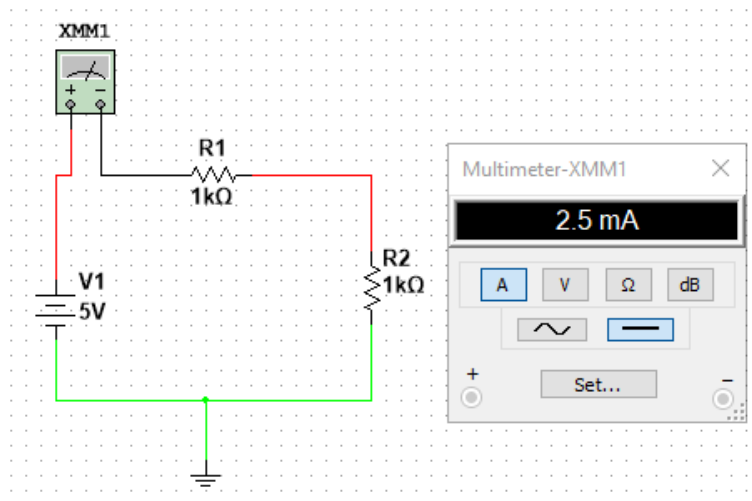


Figure 20

In Figures 17, 18, 19 and 20, the experiments analysed in the chapter "Theoretical Solution" (pages 12-14) are simulated in the simulation software "Mulitsim". In more detail, Figure 17 shows the measurement of the potential difference at the ends of the 1 KOhm resistor, where it is powered by a DC voltage source of 5 Volt. The measurement is made with the help of a multimeter set to calculate voltage. The reading of the multimeter connected in parallel to the ends of the resistor verifies the proof in the theoretical solution section (5 Volt).

Figure 18 illustrates the measurement of the current flowing through the 1 KOhm resistor. The measurement is made with the help of a multimeter set to calculate current intensity. The reading of the multimeter connected in series with the resistor verifies the proof in the theoretical solution section (5 mA). Figure 19 illustrates the measurement of the potential difference at the ends of a 1 KOhm resistor, where it is powered by a DC voltage source equal to 5 volts, connected in series with another 1 KOhm resistor. The measurement is made with the aid of a multimeter set to calculate voltage. The reading of the multimeter, which is connected in parallel to the ends of the resistor, verifies the proof in the theoretical solution section (2,5 Volt).

Figure 20 illustrates the measurement of the current flowing through the 1 KOhm resistor, connected in series with another 1 KOhm resistor. The measurement is made with the aid of a multimeter set to calculate current intensity. The reading of the multimeter connected in series with the resistors verifies the proof in the theoretical solution chapter (2,5 mA).

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Experimental Solution

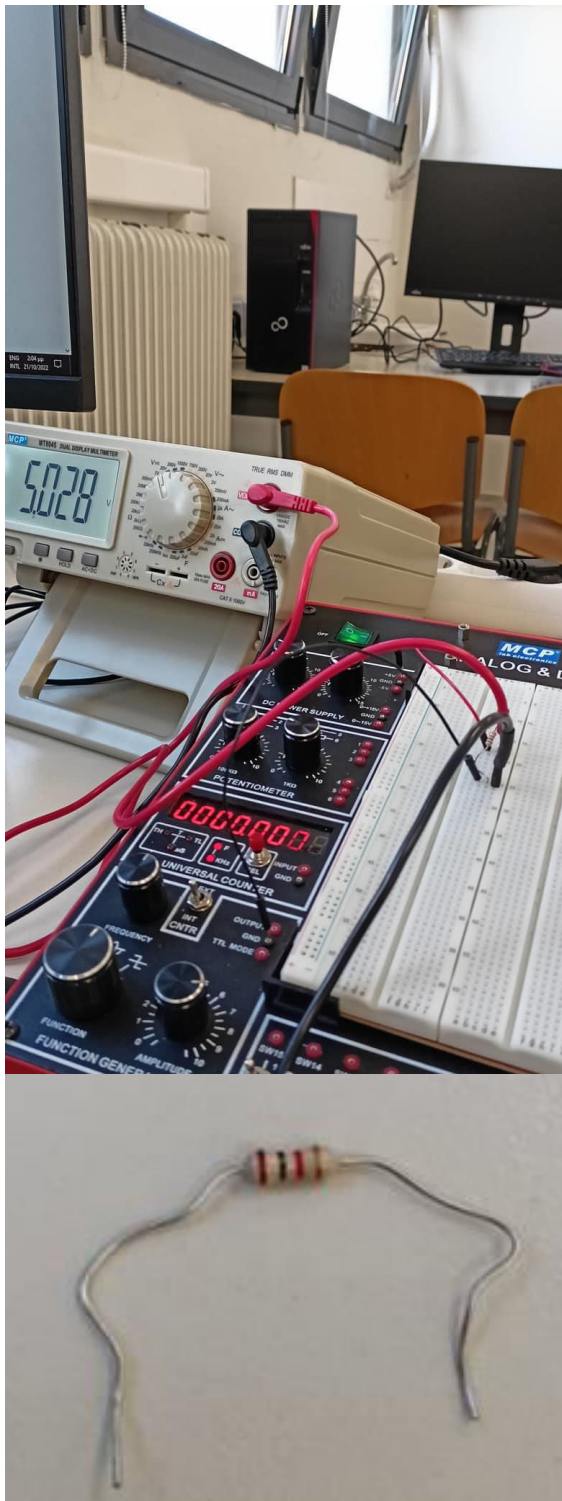


Figure 21

Resistance 1 KOhm

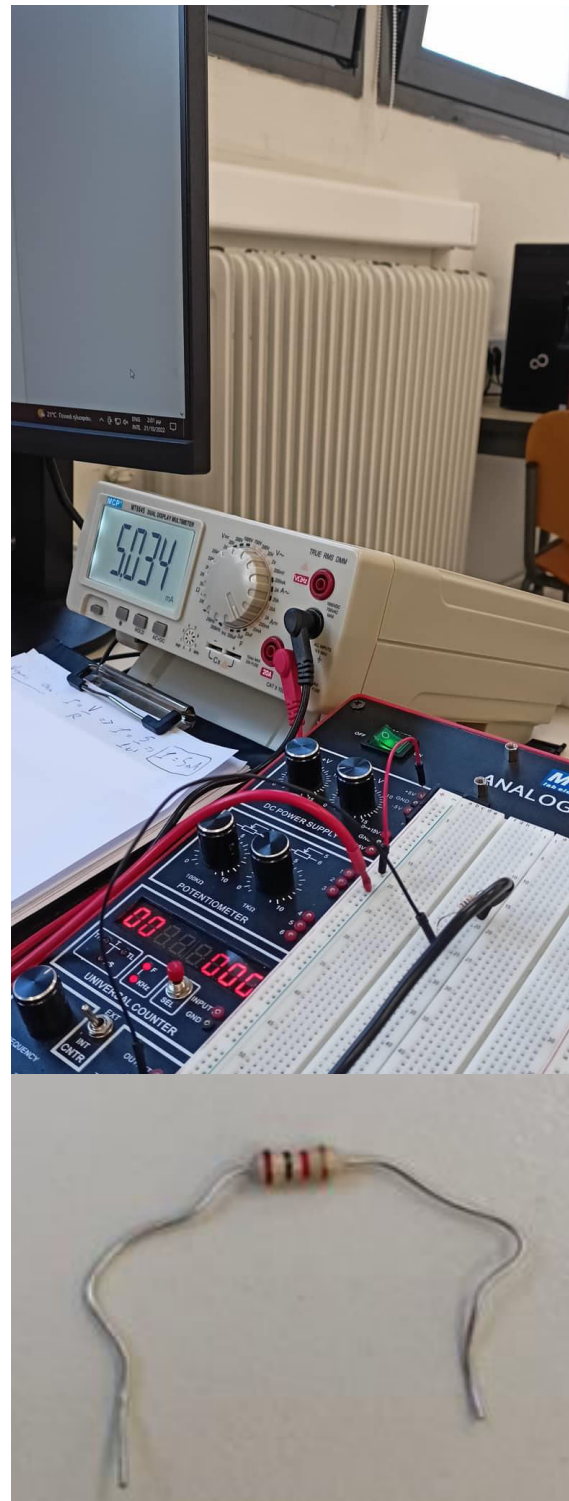


Figure 22

For the experimental solution carried out in the laboratory environment, the interface board "Breadboard M21-7000A", a digital multimeter bench "MT8045" and a 1 KOhm resistor were used.

Measuring potential difference - Figure 21

First, we take a cable and connect the DC voltage source equal to 5 Volt to a hole on the board. Next, we connect the ends of the resistor with a resistance of 1 KOhm on the board, to holes that are not the same point, where the same point is a horizontal five-hole array. Then, after setting the digital multimeter to measure voltage and to be on a larger scale than the reading we expect it to give (20 V), we connect the terminals in parallel with the resistor. We complete the circuit assembly with another wire connecting the ground to the circuit. We turn off the power switch on the board and observe that the multimeter reading is 5.028 volts by verifying the proofs in "Theoretical Solution" (pages 12-14) and "Simulation Solution" (pages 15-16). The slight difference in reading from the other two chapters is due to the tolerance of the resistor. This is because the resistor has a tolerance of -5%, i.e., -5% deviation from its true value. According to the colour code of resistors, the gold 4th stripe on the resistor indicates a $\pm 5\%$ tolerance tolerance (see Table 1 page 10). The procedure described is illustrated in Figure 21.

Measuring current intensity - Figure 22

The diode in Figure 22, starts by taking a cable and connecting the DC source of 5 Volt DC voltage to a hole on the board. Next, we connect the ends of the resistor with a 1 KOhm resistor onto the board, in holes that are not the same point, where the same point is a horizontal five-hole pattern. Then, after setting the digital multimeter to measure current intensity and to be on a larger scale than the reading we expect it to give (20 mA), we connect the terminals in series with the resistor. We complete the circuit assembly with another wire connecting the ground to the circuit. The positive terminal of the multimeter is connected to the shorted holes, while the negative terminal is connected to the shorted holes at one end of the resistor. We turn off the power switch on the board and observe that the multimeter reading is 5.034 mA by verifying the proofs in "Theoretical Solution" (pages 12-14) and "Simulation Solution" (pages 15-16). The slight difference in reading from the other two chapters is due to the tolerance of the resistor. This is because the resistor has a tolerance of -5%, i.e., -5% deviation from its true value. According to the colour code of resistors, the gold 4th stripe on the resistor indicates a $\pm 5\%$ tolerance tolerance (see Table 1 page 10).

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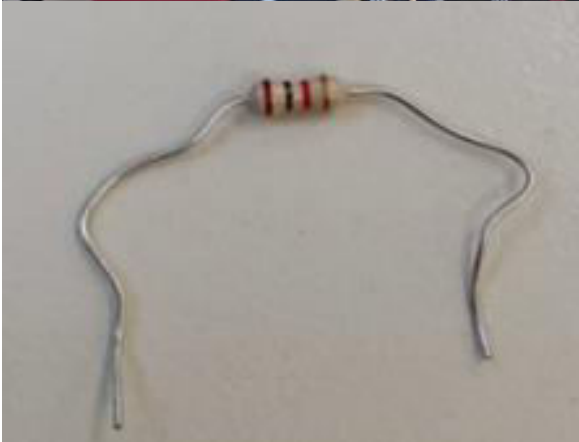
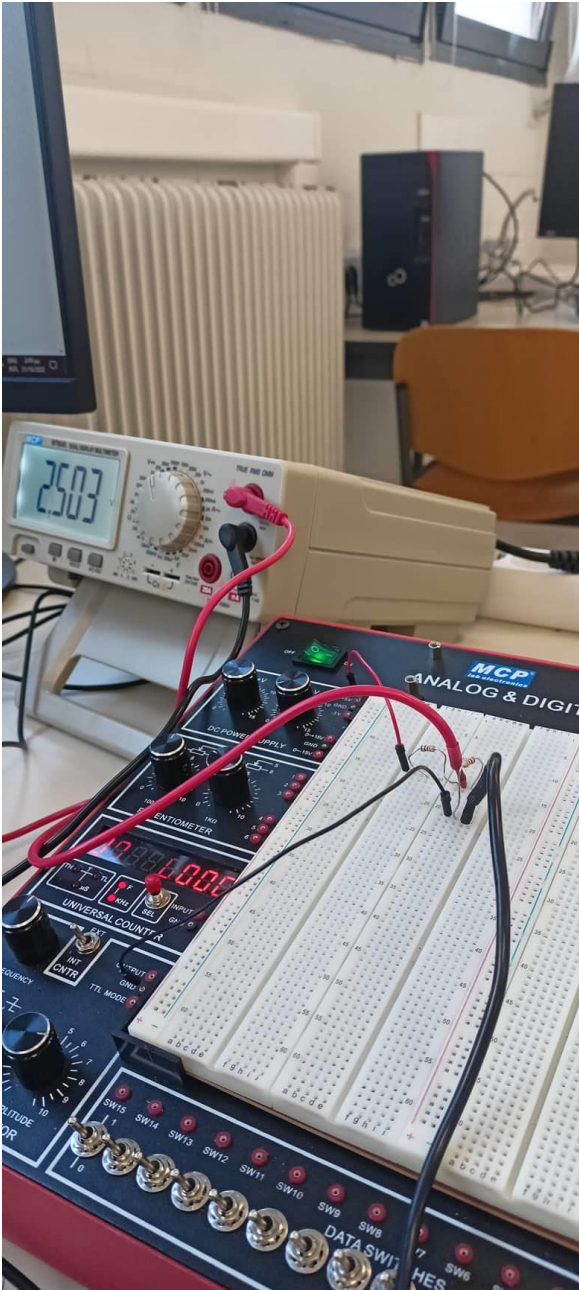


Figure 23

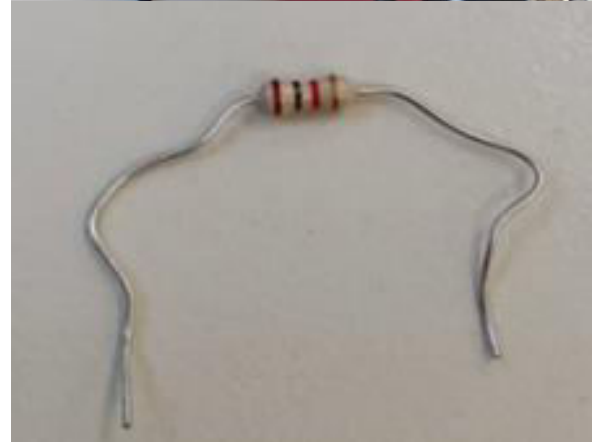
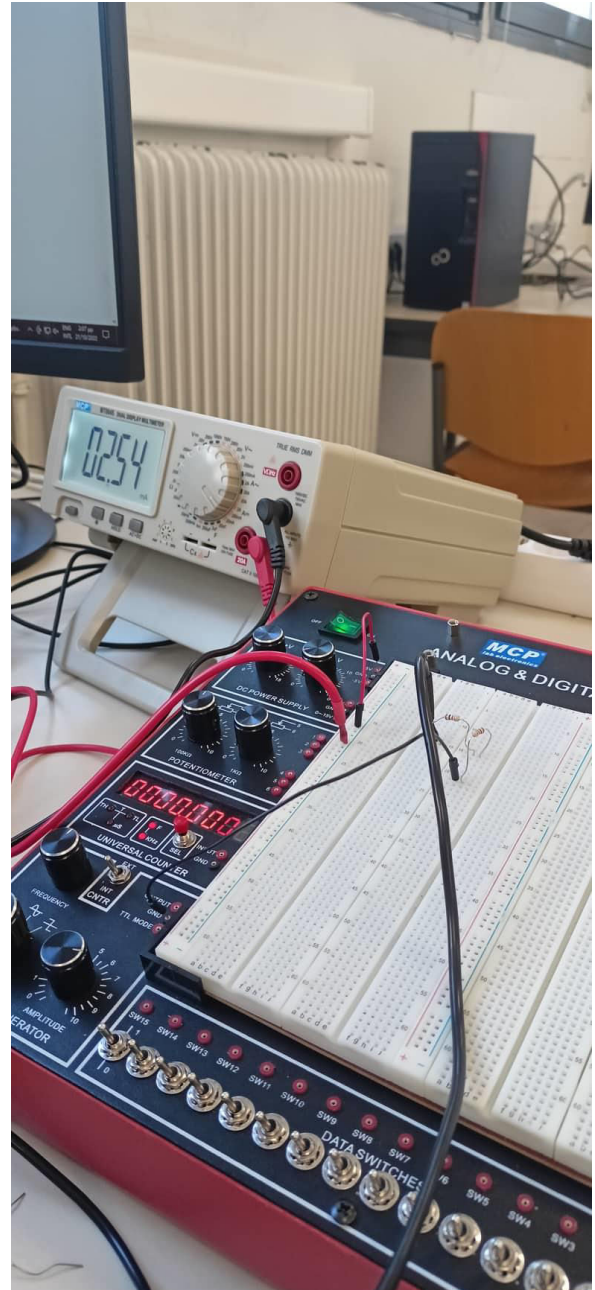


Figure 24

Resistance 2 KOhm

For the experimental solution carried out in the laboratory environment, the interface board "Breadboard M21-7000A", a digital multimeter bench "MT8045" and two 1 KOhm resistors were used.

Measuring potential difference - Figure 23

First, we take a cable and connect the DC voltage source equal to 5 Volt to a hole on the board. Next, we connect the ends of the resistor with a resistance of 1 KOhm on the board, to holes that are not the same point, where the same point is a horizontal five-hole array. Repeat the same for the other resistor and connect it in series with the other one. Then, after setting the digital multimeter to measure voltage and to be on a larger scale than the reading we expect it to give (20 V), we connect the terminals in parallel with one resistor. We complete the circuit assembly with another wire connecting the ground to the circuit. We turn off the power switch on the board and observe that the multimeter reading is 2.503 volts by verifying the proofs in the "Theoretical Solution" (pages 12-14) and "Simulation Solution" (pages 14-15) chapters. The slight difference in reading from the other two chapters is due to the tolerance of the resistor. This is because the resistor has a tolerance of -5%, i.e., -5% deviation from its true value. According to the colour code of resistors, the gold 4th stripe on the resistor indicates a $\pm 5\%$ tolerance (see Table 1 page 10). The procedure described is illustrated in Figure 23.

Measuring current intensity - Figure 24

The diode in Figure 24, starts by taking a cable and connecting the DC source of 5 Volt DC voltage to a hole on the board. Next, we connect the ends of the resistor with a 1 KOhm resistor onto the board, in holes that are not the same point, where the same point is a horizontal five-hole array. Repeat the same for the other resistor and connect it in series with the other one. Then, after setting the digital multimeter to measure current intensity and to be on a larger scale than the reading we expect it to give (20 mA), we connect the terminals in series with the resistors. We complete the circuit assembly with another wire connecting the ground to the circuit. The positive terminal of the multimeter is connected to the shorted holes, while the negative terminal is connected to the shorted holes at one end of the resistor. We turn off the power switch on the board and observe that the multimeter reading is 0.254 mA by verifying the proofs in "Theoretical Solution" (pages 12-14) and "Simulated Solution" (pages 14-15). The slight difference in reading from the other two chapters is due to the tolerance of the resistor. This is because the resistor has a tolerance of -5%, i.e., -5% deviation from its true value. According to the colour code of resistors, the gold 4th stripe on the resistor indicates a $\pm 5\%$ tolerance (see Table 1 page 10).

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3. Measurement of AC A.C. / Measurement of period and calculation of frequency F from it

General

In the chapters "Simulated Solution" and "Experimental Solution" the components used and the observations made during the experiment are presented in detail with snapshots from the "Multisim" software and photos of the experiment from the laboratory environment respectively, while in the chapter "Theoretical Solution" the mathematical proof of the experiment is analysed.

Theoretical Solving

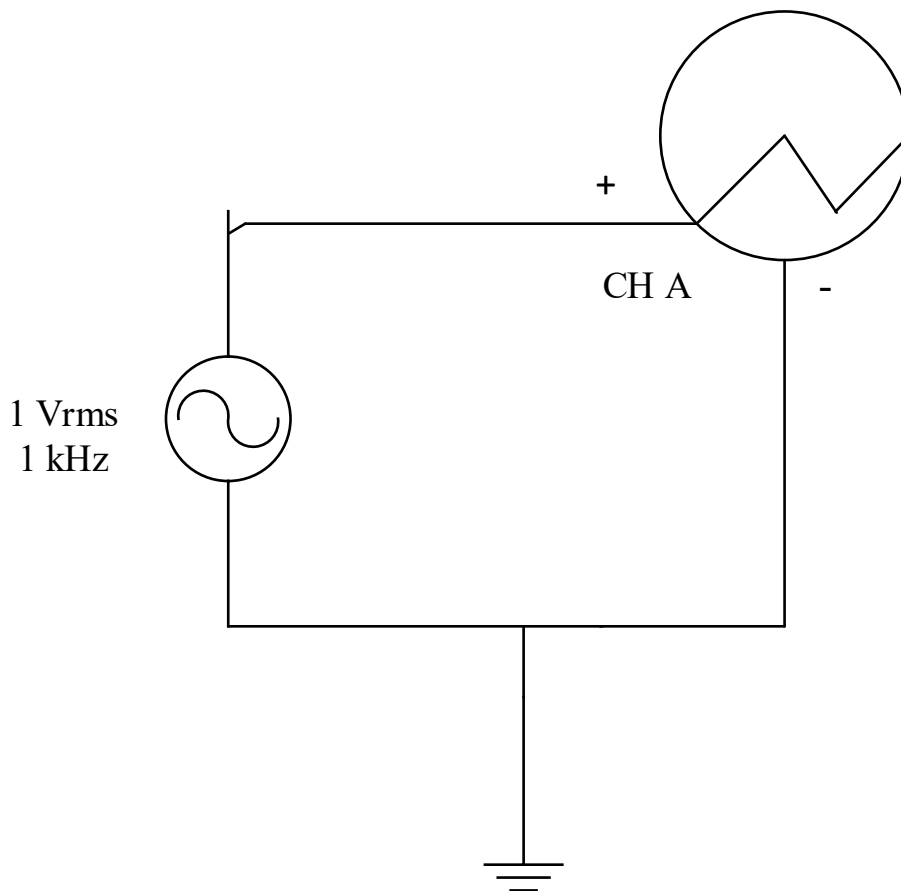


Figure 25

Figure 25 shows a schematic illustration of the connection of a channel of the oscilloscope to a signal emitted by an AC voltage source A.C. The signal generator is set at a frequency of 1000 Hz and the experiment is carried out for various input signals (Vrms) in Volts.

The voltage at A.C. does not remain constant with respect to time unlike the voltage at D.C. In fact, it assumes a sinusoidal form that varies over time as shown in the graph in Figure 26.

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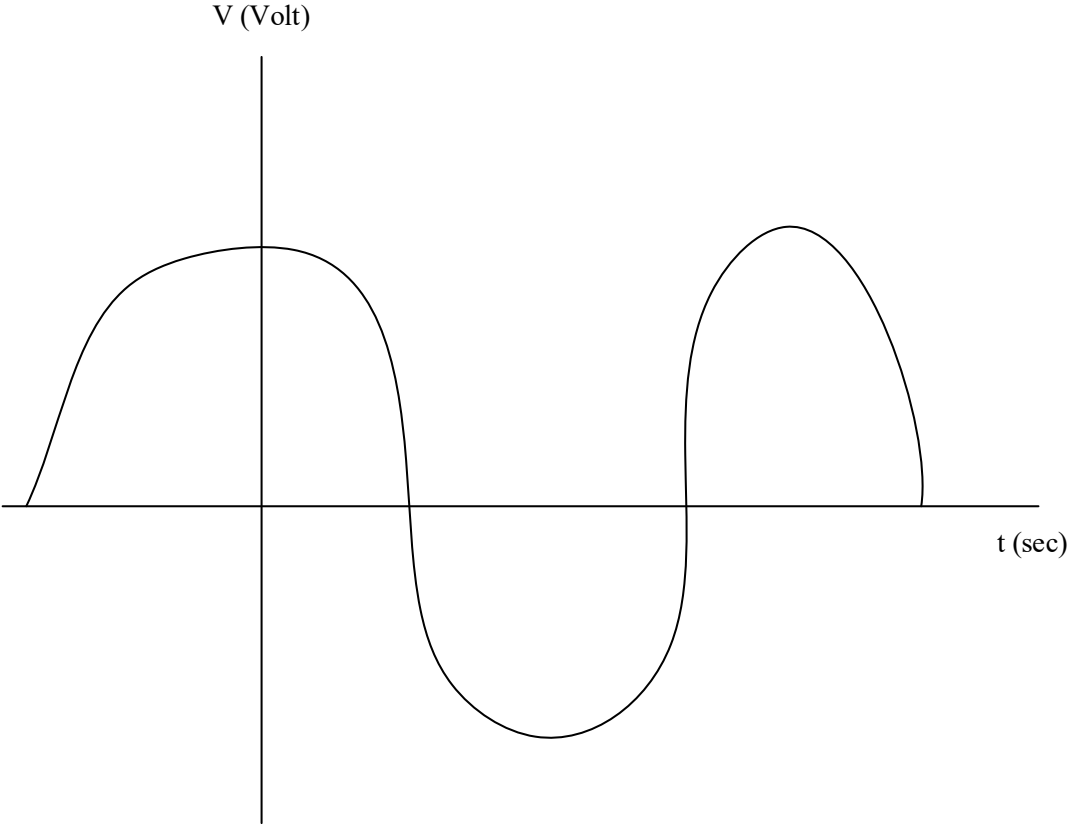


Figure 26

Simulative Resolution

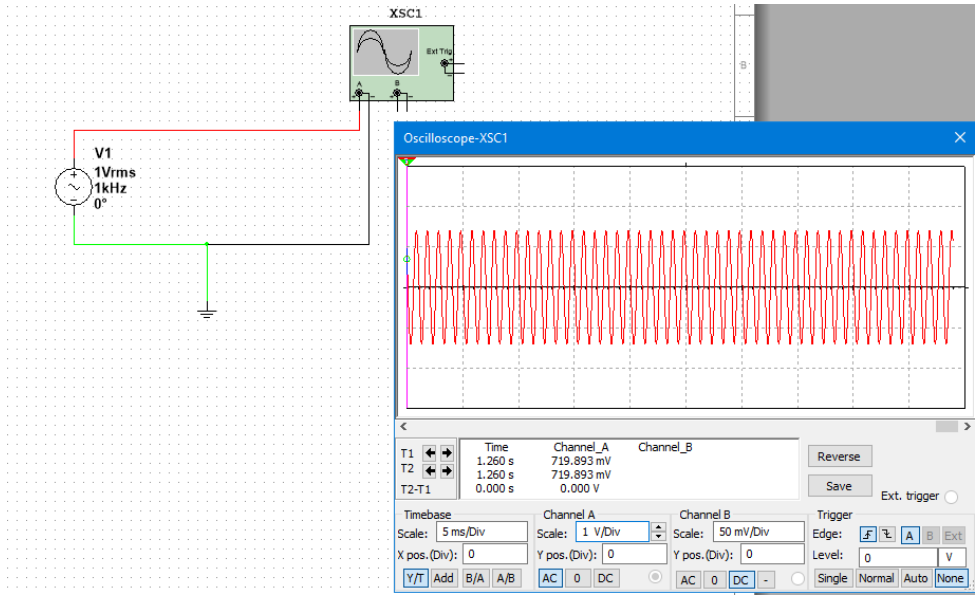


Figure 27

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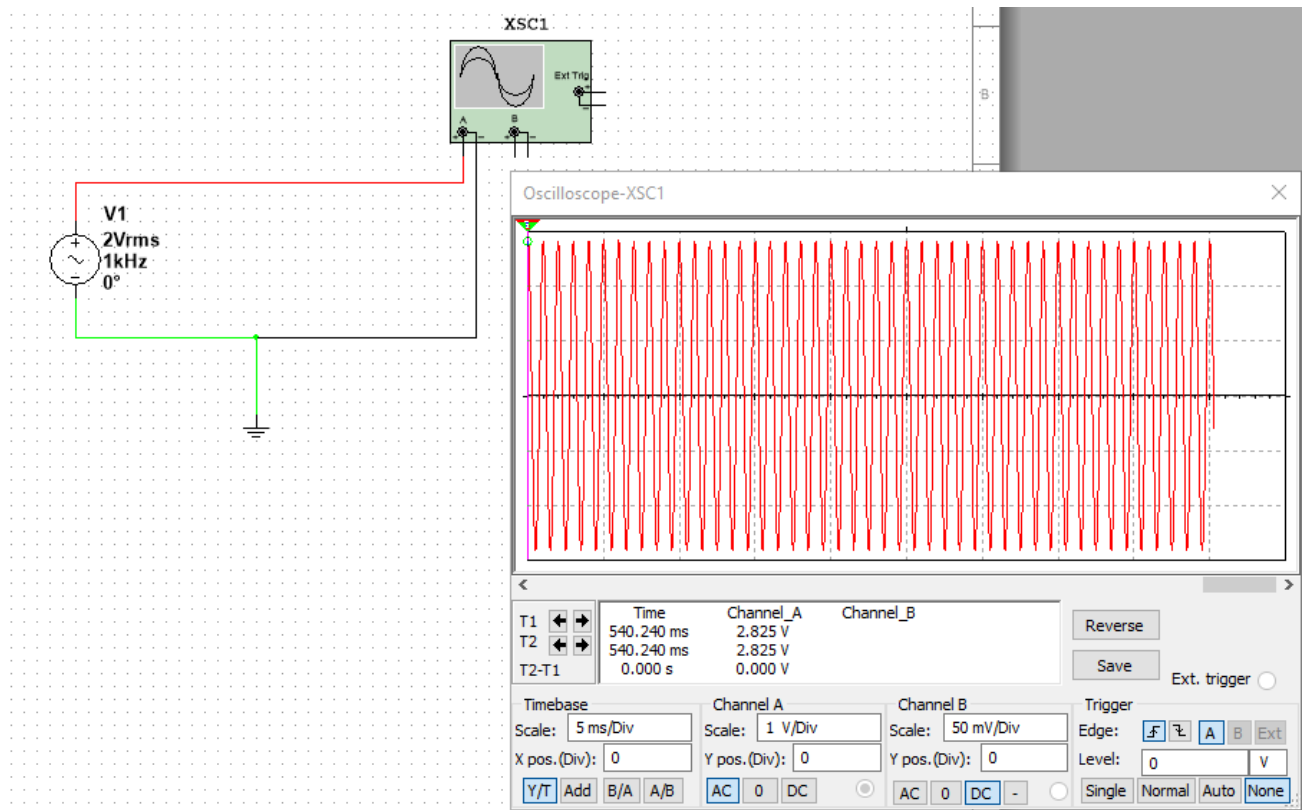


Figure 28

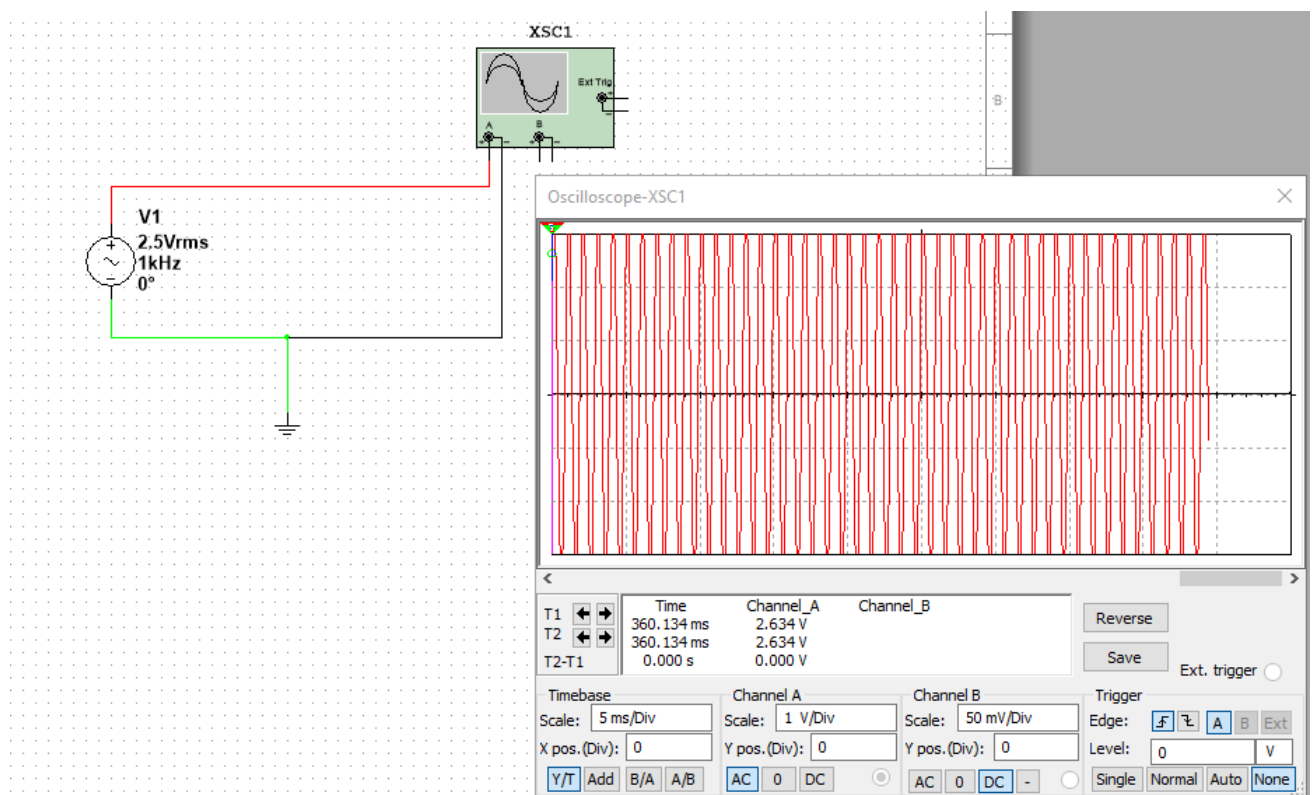


Figure 29

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Input signal active value in Volt	Voltage value Vp-p (peak to peak) in Volt	Calculation of voltage value Vp in Volt $V_p = V_{p-p} / 2$	Calculation of active value Ven in Volt $V_{en} = V_p / \sqrt{2}$
1	3	1.5	1.06
2	5.5	2.75	1.9
2.5	6	3	2.1

Table 4

In Figures 27, 28, 29 are simulated in the simulation software "Multisim", the experiments analyzed in the chapter "Theoretical Solution" (pages 21-22) for the measurement of alternating voltage A.C. The data recorded in Table 4 are obtained from the calculations performed on the oscilloscope screen and on the sinusoidal signal it emits. More specifically, V_{p-p} is calculated from the product of: τετραγωνάκια $\times (\frac{\text{Volt}}{\text{Div}})$. V_p from the quotient $\frac{V_{p-p}}{2}$ and V_{ev} by the quotient $\frac{V_p}{\sqrt{2}}$.

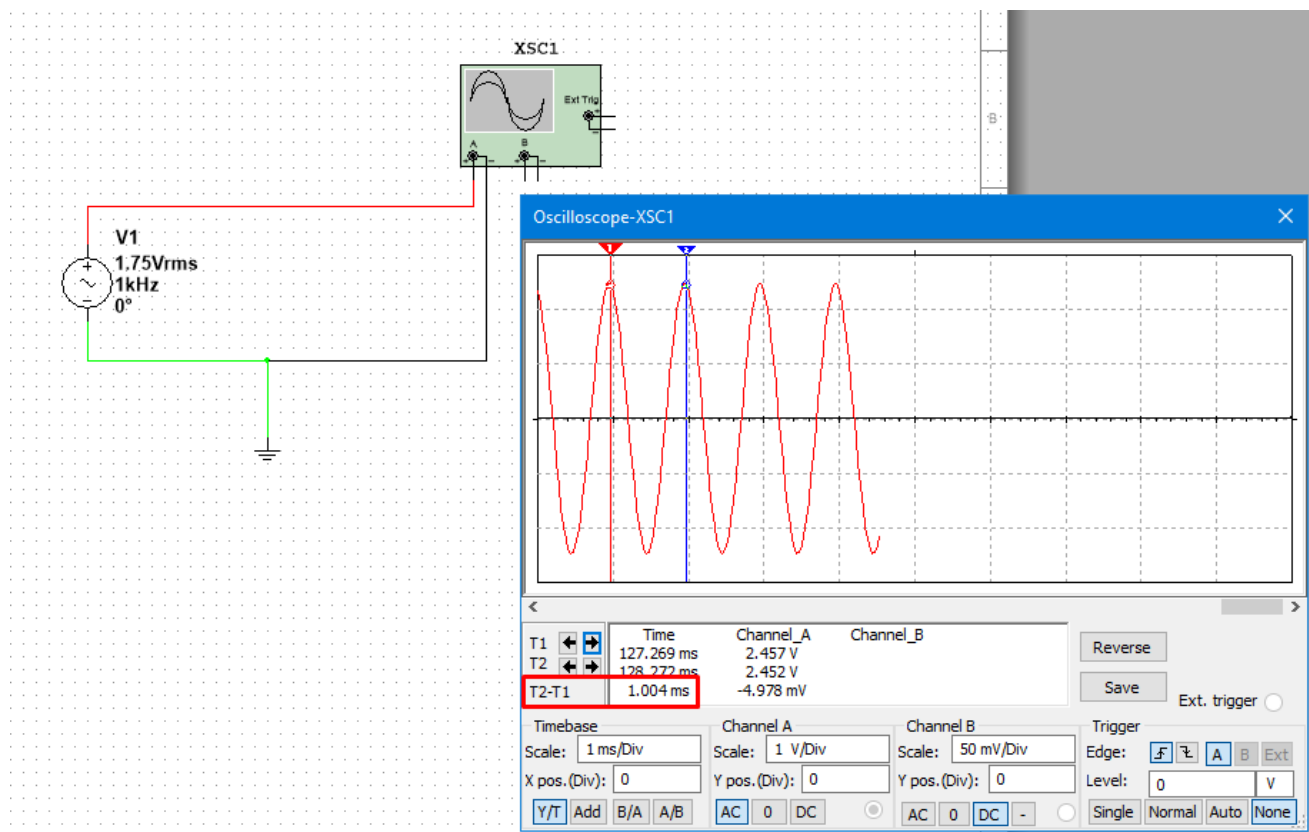


Figure 30

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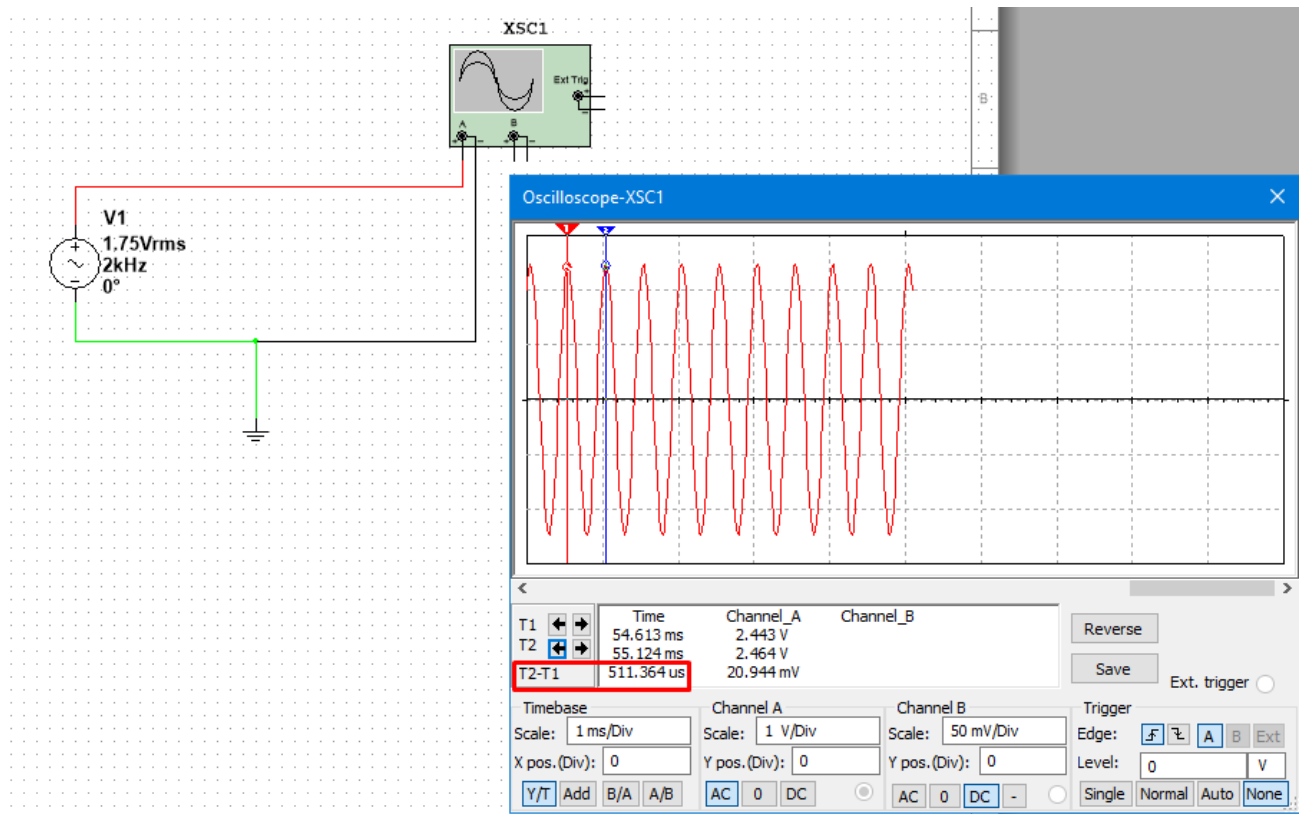


Figure 31

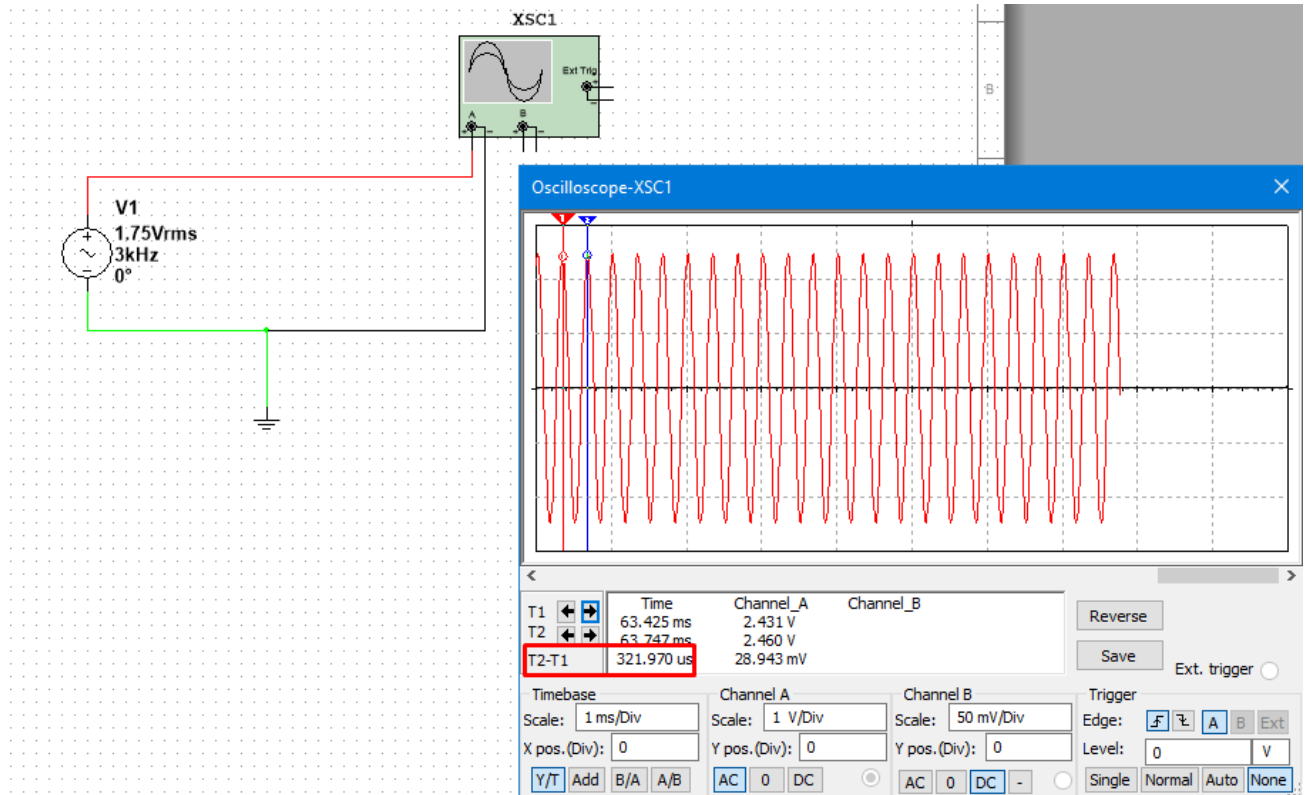


Figure 32

ELECTRONICS

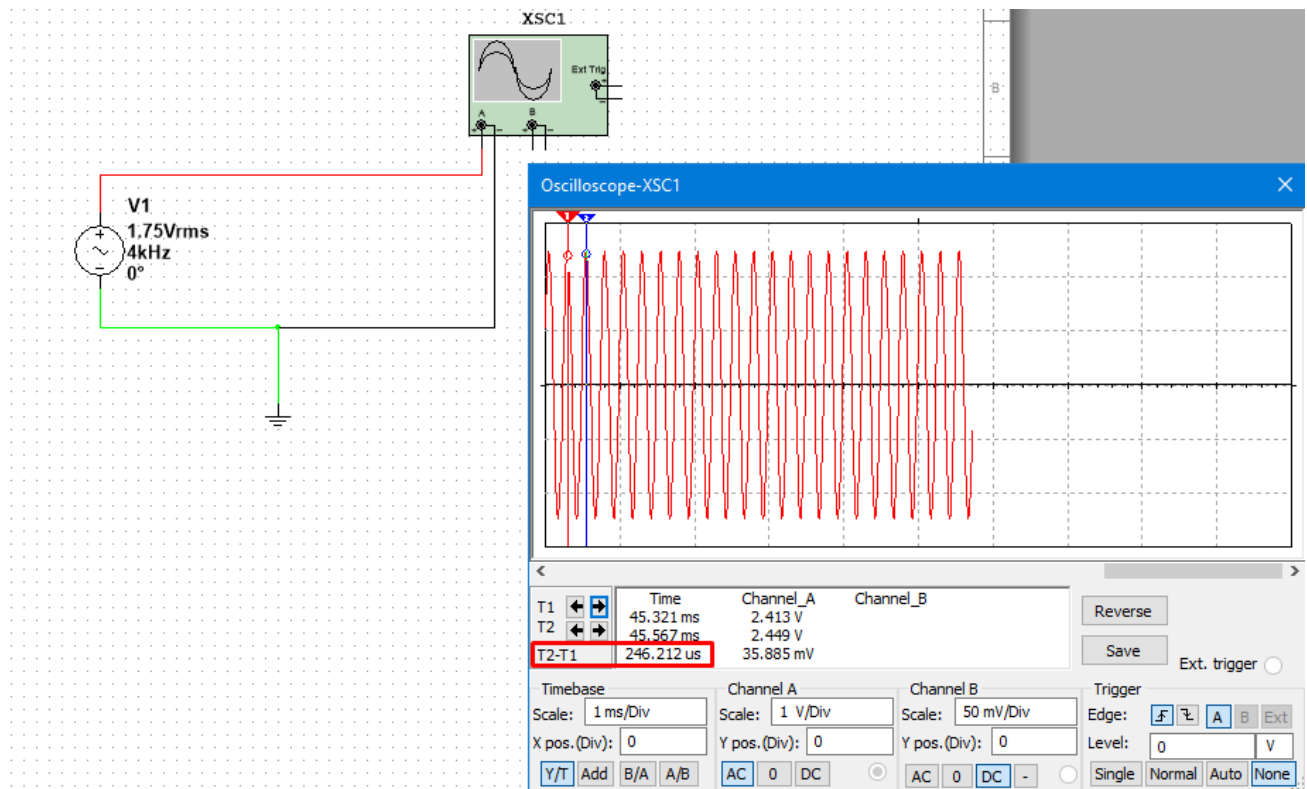


Figure 33

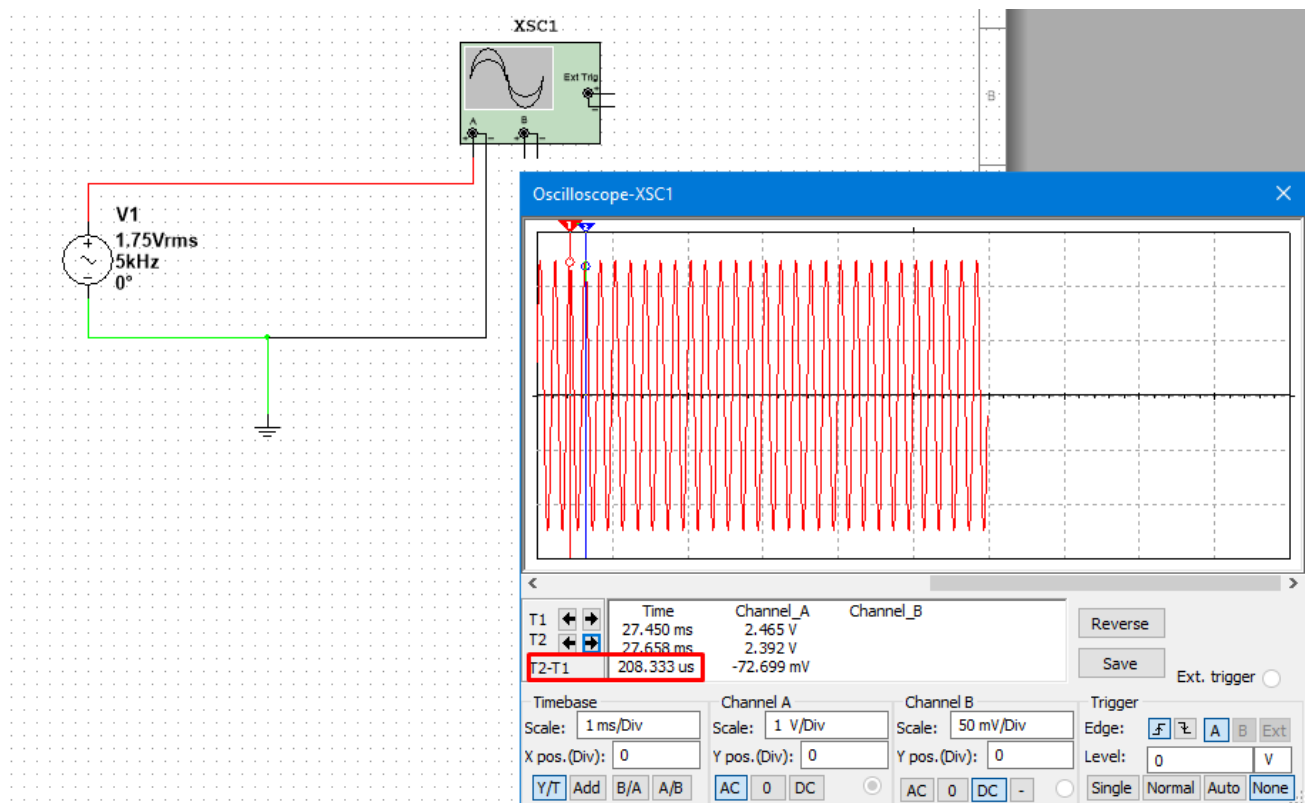


Figure 34

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Frequency f / Hz	T period / msec	Frequency calculation Hz
1000	1	1000
2000	0.5	2000
3000	0.3	3333,3
4000	0.24	4166,7
5000	0.2	5000

Table 5

Figures 30, 31, 32, 33 and 34 show simulated experiments in the simulation software "Multisim" for calculating the period and frequency of AC voltage A.C. The period is calculated from the formula $\text{Time} \times \left(\frac{\text{Time}}{\text{Div}} \right)$, while the frequency is given by the formula $f = \frac{1}{T}$.

Experimental Solution

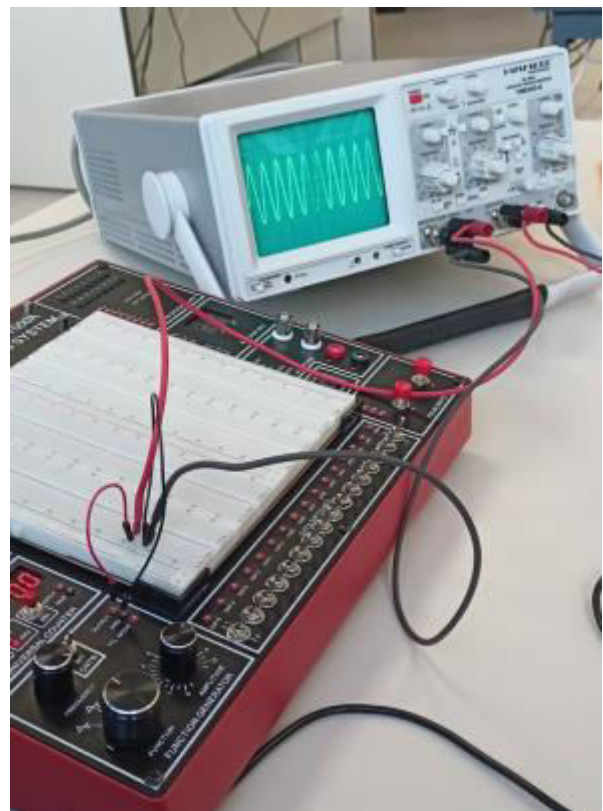
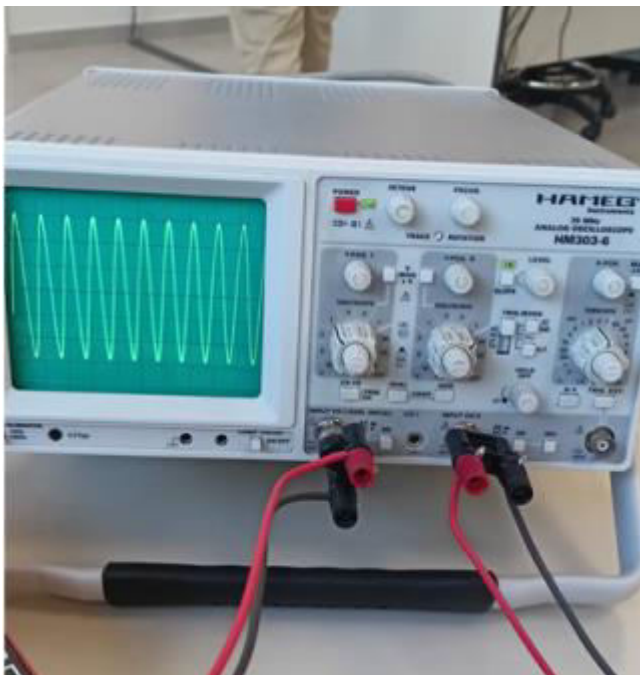


Figure 35 Figure 36

ELECTRONICS

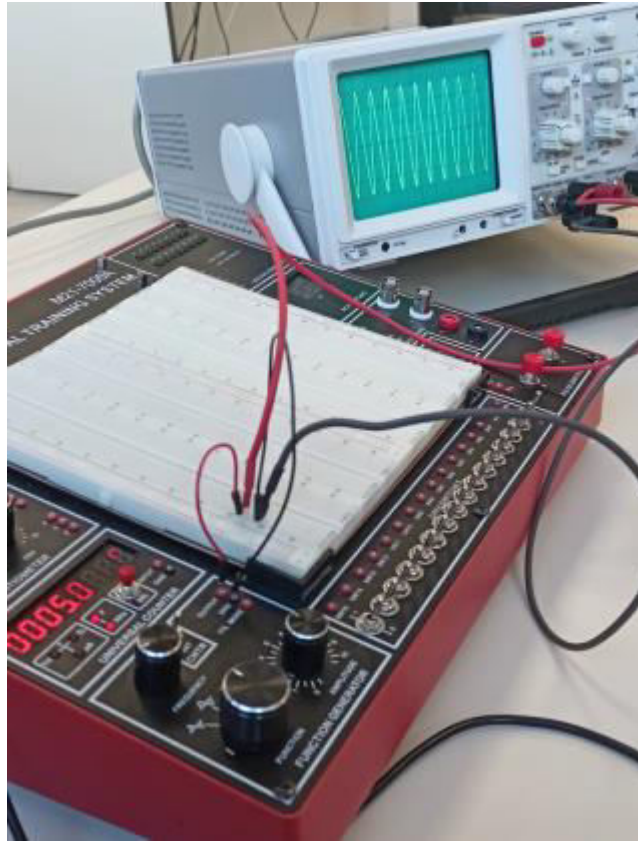


Figure 37

For the experimental solution carried out in the laboratory environment, the interface board "Breadboard M21-7000A", the oscilloscope "HAMEG HM203-5" and two cables were used. Initially, we set the signal frequency to 1 KHz (Figure 36) and later to 5 KHz (Figure 35, Figure 37). Then, with one cable we connect the input signal to one hole (AC voltage source in A.C.) and with the other cable we connect the ground to another hole. Next, we take the two terminals of one channel (A or B) of the oscilloscope and connect them to the circuit. The positive terminal (red) is connected to the source and the negative terminal is connected to the ground (black). Aligning the x and y axes so that they are exactly centered on the oscilloscope screen, we set the Volts/Div and Time/Div switches to 1 Volt/Div and 1 ms/sec respectively. We observe that on the oscilloscope screen, the data presented by the waveform (V_{p-p} , V_p , T), verify the results in the "Simulation Solution" chapter (pages 21-26).

4. Questions

Question 1

What is the value of the DC voltage that causes a 3-square dot divergence and the Volt/DIV switch is at position 2?

$$V = 3 \text{ squares} \times 2 \frac{\text{Volt}}{\text{Div}} \rightarrow V = 6 V_{p-p}$$

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Question 2

What are the values of the injected voltage V_{p-p} , V_p and $V_{\epsilon v}$, which causes the dot deviation of 0.5, 2 and 3 squares respectively from peak to peak and the Volt/DIV switch is at 50 mV, 2 V and 5 V respectively?

0.5 squares, 50 mV

$$V_{p-p} = 0.5 \text{ squares} \times 50 \text{ mV} \rightarrow V_{p-p} = 25 \text{ mV}$$

$$V_p = \frac{V_{p-p}}{2} \rightarrow V_p = \frac{25 \text{ mV}}{2} \rightarrow V_p = 12.5 \text{ mV}$$

$$V_{\epsilon v} = \frac{V_p}{\sqrt{2}} \rightarrow V_{\epsilon v} = \frac{12.5 \text{ mV}}{\sqrt{2}} \rightarrow V_{\epsilon v} = 8.8 \text{ mV}$$

2 squares, 2 V

$$V_{p-p} = 2 \text{ squares} \times 2 \text{ V} \rightarrow V_{p-p} = 4 \text{ V}$$

$$V_p = \frac{V_{p-p}}{2} \rightarrow V_p = \frac{4 \text{ V}}{2} \rightarrow V_p = 2 \text{ V}$$

$$V_{\epsilon v} = \frac{V_p}{\sqrt{2}} \rightarrow V_{\epsilon v} = \frac{2 \text{ V}}{\sqrt{2}} \rightarrow V_{\epsilon v} = 1.4 \text{ V}$$

3 squares, 5 V

$$V_{p-p} = 3 \text{ squares} \times 5 \text{ V} \rightarrow V_{p-p} = 15 \text{ V}$$

$$V_p = \frac{V_{p-p}}{2} \rightarrow V_p = \frac{15 \text{ V}}{2} \rightarrow V_p = 7.5 \text{ V}$$

$$V_{\epsilon v} = \frac{V_p}{\sqrt{2}} \rightarrow V_{\epsilon v} = \frac{7.5 \text{ V}}{\sqrt{2}} \rightarrow V_{\epsilon v} = 5.3 \text{ V}$$

Question 3

What is the frequency of the AC voltage when the peak-to-peak deviation is 0.5, 2 and 3 squares and the Time/DIV switch is set to 2, 5 and 50 ms respectively?

0.5 squares, 2 ms

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$$T = 0.5 \text{ squares} \times 2 \text{ ms} \rightarrow T = 1 \text{ ms}$$

$$f = \frac{1}{T} \rightarrow f = \frac{1}{10^{-3}} \rightarrow f = 1000 \text{ Hz}$$

2 squares, 5 ms

$$T = 2 \text{ squares} \times 5 \text{ ms} \rightarrow T = 10 \text{ ms}$$

$$f = \frac{1}{T} \rightarrow f = \frac{1}{10^{-3} \times 10} \rightarrow f = 100 \text{ Hz}$$

3 squares, 50 ms

$$T = 3 \text{ squares} \times 50 \text{ ms} \rightarrow T = 150 \text{ ms}$$

$$f = \frac{1}{T} \rightarrow f = \frac{1}{10^{-3} \times 150} \rightarrow f = 6.6 \text{ Hz}$$

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Thank you for your attention.

